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AI Cost & Tokenomics Guide

Budget templates, cost comparison tables, and cost optimization strategies

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1. Executive Summary

AI infrastructure costs are driven by multiple variables: model complexity, input/output volume, compute requirements, and operational overhead. This guide helps organizations build accurate cost models, benchmark against peers, and identify optimization opportunities.

2. Token Pricing Models

LLM Token Economics

Prompt Tokens: Cost per 1K tokens for model input (typically lower rate) **Completion Tokens:** Cost per 1K tokens for model output (typically 2-4x higher rate) **Example Pricing (as of 2024):** GPT-4 Turbo: $10/1M prompt tokens, $30/1M completion tokens GPT-3.5: $0.50/1M prompt tokens, $1.50/1M completion tokens

Fine-Tuning Economics

Training Cost: Per token during fine-tuning phase (one-time) **Inference Cost:** Discounted rate for using fine-tuned models (ongoing) **Storage:** Cost per model stored (usually $0.01-0.05/day per model)

3. Cost Components

Direct Infrastructure Costs

Component	Cost Driver	Monthly Cost Range
LLM API calls	Tokens processed	$100 - $50K
Fine-tuning	Hours + tokens	$500 - $20K
Vector database	Storage & queries	$50 - $5K
Compute (GPUs)	Hours used	$500 - $30K
Storage & CDN	Data volume	$100 - $5K

Operational Costs

• Data preparation & labeling: $2K-20K/month • Model monitoring & optimization: $1K-10K/month • Security & compliance tools: $500-5K/month • Support & maintenance: $1K-15K/month

4. Budget Template

Annual AI Budget Forecast

Cost Category	Q1	Q2	Q3	Q4	Annual
LLM API Calls	$5K	$8K	$12K	$15K	$40K
Compute & Infrastructure	$10K	$10K	$10K	$10K	$40K
Data & Training	$5K	$3K	$2K	$2K	$12K
Team & Operations	$20K	$20K	$20K	$20K	$80K
Tools & Monitoring	$2K	$2K	$2K	$2K	$8K
TOTAL	**$42K**	**$43K**	**$46K**	**$49K**	**$180K**

5. Cost Optimization Strategies

Token Reduction Techniques

Prompt Optimization: Reduce prompt size through template engineering (5-20% savings) **Caching:** Cache repeated prompts and system instructions (10-30% savings) **Batching:** Process multiple requests together instead of individually (15-25% savings) **Model Selection:** Use smaller models where appropriate (50-80% savings vs GPT-4)

Infrastructure Optimization

Reserved Capacity: Commit to usage for 1-3 year discounts (20-40% savings) **Spot Instances:** Use interruptible compute for non-critical workloads (60-80% savings) **Local Models:** Deploy open-source models for cost-sensitive use cases (90%+ savings)

6. Benchmarking

Industry Benchmarks (2024)

Use Case	Tokens/Month	Monthly Cost	Cost/Transaction
Customer Support Chatbot	100M	$3K	$0.03
Content Generation	50M	$1.5K	$0.15
Data Analysis	10M	$0.5K	$0.50
Document Processing	200M	$5K	$0.05

7. Financial Forecasting

3-Year Cost Projection

Year	Monthly Avg	Annual Total	YoY Growth	Cost/User
2024	$20K	$240K	N/A	$2
2025	$18K	$216K	-10%	$1.80
2026	$22K	$264K	+22%	$2.20

Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

We're here to help.

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For consultation on implementation, governance, and strategic deployment of AI initiatives.